

# Yosuke HIGUCHI

☎ (+81) 80-4118-2106 | ✉ higuchi@pcl.cs.waseda.ac.jp | 📷 YosukeHiguchi

## Education

### Waseda University

*M.S. in Computer Science and Engineering*

- Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
- Research Field: Speech Recognition, Pattern Recognition, Machine Learning

Tokyo, Japan

Apr. 2019 - Present

### Waseda University

*B.E. in Computer Science and Engineering*

- Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
- Research Field: Speech Recognition, Pattern Recognition, Machine Learning

Tokyo, Japan

Apr. 2015 - Mar. 2019

## Research Topic

### Zero-resource ASR

*Supervisor: Naohiro Tawara, Tetsuji Ogawa, Tetsunori Kobayashi*

- Proposed a framework for extracting speaker-invariant features from zero-resource languages. The framework consists of two phases: unsupervised clustering to obtain phonetic information and domain-adversarial training to suppress a variety in speakers.

Waseda University

Jun. 2018 - Mar. 2019

### Non-parallel many-to-many voice conversions with StarGAN

*Supervisor: Hirokazu Kameoka*

- Worked on improving StarGAN-based voice conversion model [Kameoka+, 2018]. New network structures and objective functions were designed to improve the model.

NTT CS Lab.

Aug. 2018 - Sept. 2018

### Extraction of speaker representations for speaker diarization

*Supervisor: Masayuki Suzuki, Gakuto Kurata*

- Worked on building a speaker model for speaker diarization. DNN-based speaker embeddings were trained to incorporate differences in acoustic conditions between recording environments.

IBM Research AI

Jun. 2019 - Oct. 2019

### End-to-end speech recognition

*Supervisor: Shinji Watanabe*

- Working on non-autoregressive end-to-end speech recognition.

JHU, Waseda Univ.

Dec. 2019 - Present

## Work Experience

### InfoDeliver Corporation

*Part-time Software Engineer*

- Involved in developing and maintaining web services and mobile apps.

Tokyo, Japan

Sept. 2016 - Jun. 2017

### NTT Communication Science Laboratories, NTT Corporation

*Research Internship*

- Research Topic: voice conversion, generative adversarial network

Kanagawa, Japan

Aug. 2018 - Sept. 2018

### IBM Research AI

*Research Internship*

- Research Topic: speaker diarization

Tokyo, Japan

Jun. 2019 - Oct. 2019

### Johns Hopking University

*Visiting Scholar*

- Research Topic: end-to-end speech recognition

Baltimore, MD

Dec. 2019 - Mar. 2020

## Publication

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### Refereed Conference

- **Yosuke Higuchi**, Naohiro Tawara, Tetsunori Kobayashi, Tetsuji Ogawa, "Speaker Adversarial Training of DPGMM-based Feature Extractor for Zero-resource Languages", In *Proceedings of 20th Annual Conference of International Speech Communication Association (INTERSPEECH)*, 2019.
- **Yosuke Higuchi**, Masayuki Suzuki, Gakuto Kurata, "Speaker Embeddings Incorporating Acoustic Conditions for Diarization", In *Proceedings of IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020. (to appear)

### Domestic Conference

- **Yosuke Higuchi**, Naohiro Tawara, Tetsuji Ogawa, Tetsunori Kobayashi, "Speaker Invariant Feature Extraction for Zero-Resource Languages", In *2019 Spring Meeting of Acoustical Society of Japan (ASJ)*, March 2019.
- **Yosuke Higuchi**, Naohiro Tawara, Tomoharu Iwata, Atsunori Ogawa, Tetsunori Kobayashi, Tetsuji Ogawa, "Noise-robust Attention Learning for End-to-End Speech Recognition", In *2020 Spring Meeting of Acoustical Society of Japan (ASJ)*, March 2020.

### Domestic Workshop

- **Yosuke Higuchi**, Naohiro Tawara, Tetsunori Kobayashi, Tetsuji Ogawa, "Speaker Adversarial Training of DPGMM-based Feature Extractor for Zero-resource Languages", In *IPSJ SIG Technical Reports on Spoken Language Processing (IPSJ-SLP)*, September 2019. - Yahoo! JAPAN award

## Skills

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|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming language</b>    | Proficient in: Python, Bash, Pytorch, Chainer, C/C++, Java, JavaScript, LaTeX<br>Also basic ability with: MATLAB, Node.js, OCaml |
| <b>Software &amp; Tools</b>    | ESPnet, Kaldi                                                                                                                    |
| <b>Deep learning framework</b> | PyTorch, Chainer                                                                                                                 |
| <b>Natural language</b>        | Japanese (native), English (TOEFL iBT: 102, TOEIC: 950)                                                                          |